



Model based planning in Continuous-Time MDP

1. Modeling human's MB planning in CTMDP | 2. Research plan studying planning in the brain

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1. Modeling human's MB planning in CTMDP

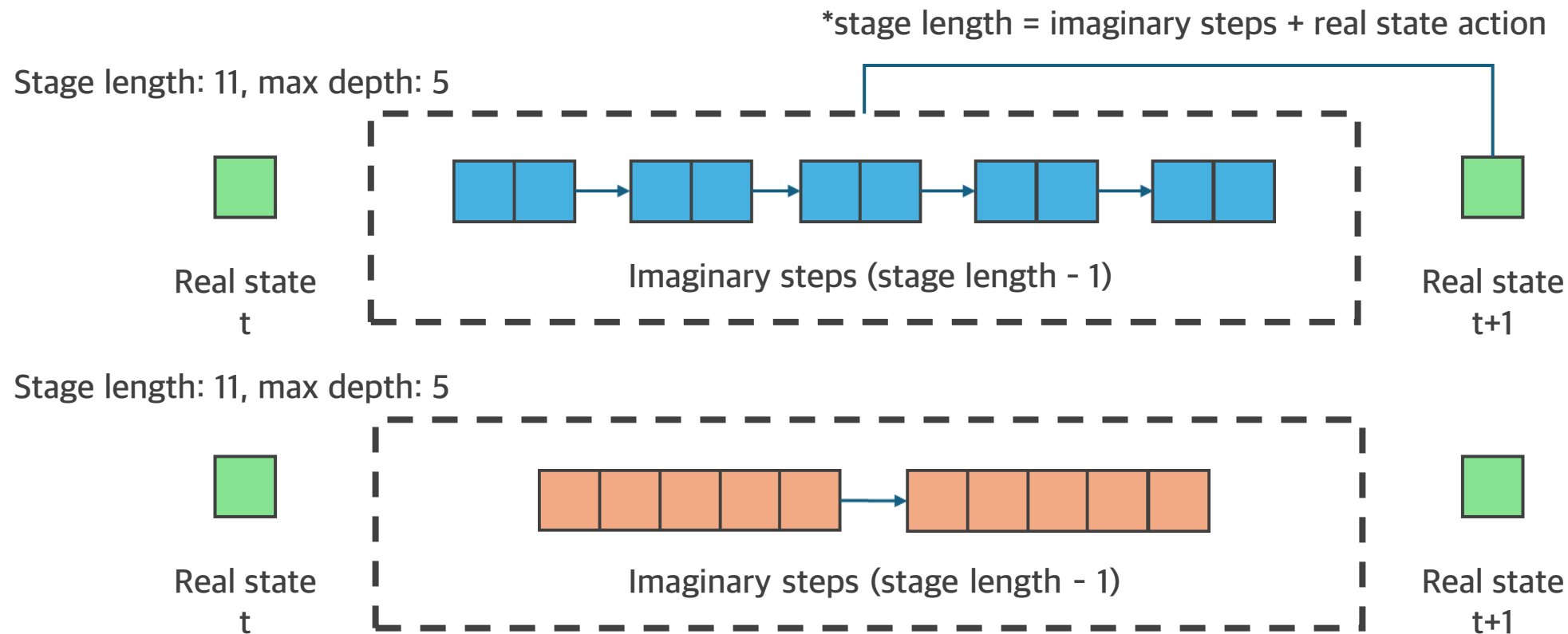
My research question is...

“How does the brain allocate “thinking time” to predict futures and execute plans in real-time games?”

- In complex games, humans can delay action to evaluate future consequences (“thinking time”), rather than acting at fixed ticks.
- Frame-by-frame gameplay provides a behavioral handle on decision time at state-level resolution.
- Naturalistic video-game fMRI is feasible, and RL models can be used as computational lenses for latent state representations and internal-model updates.
- We can test whether functional connectivity tracks when people pause and perform better.

RL agent as a proxy of model based planning

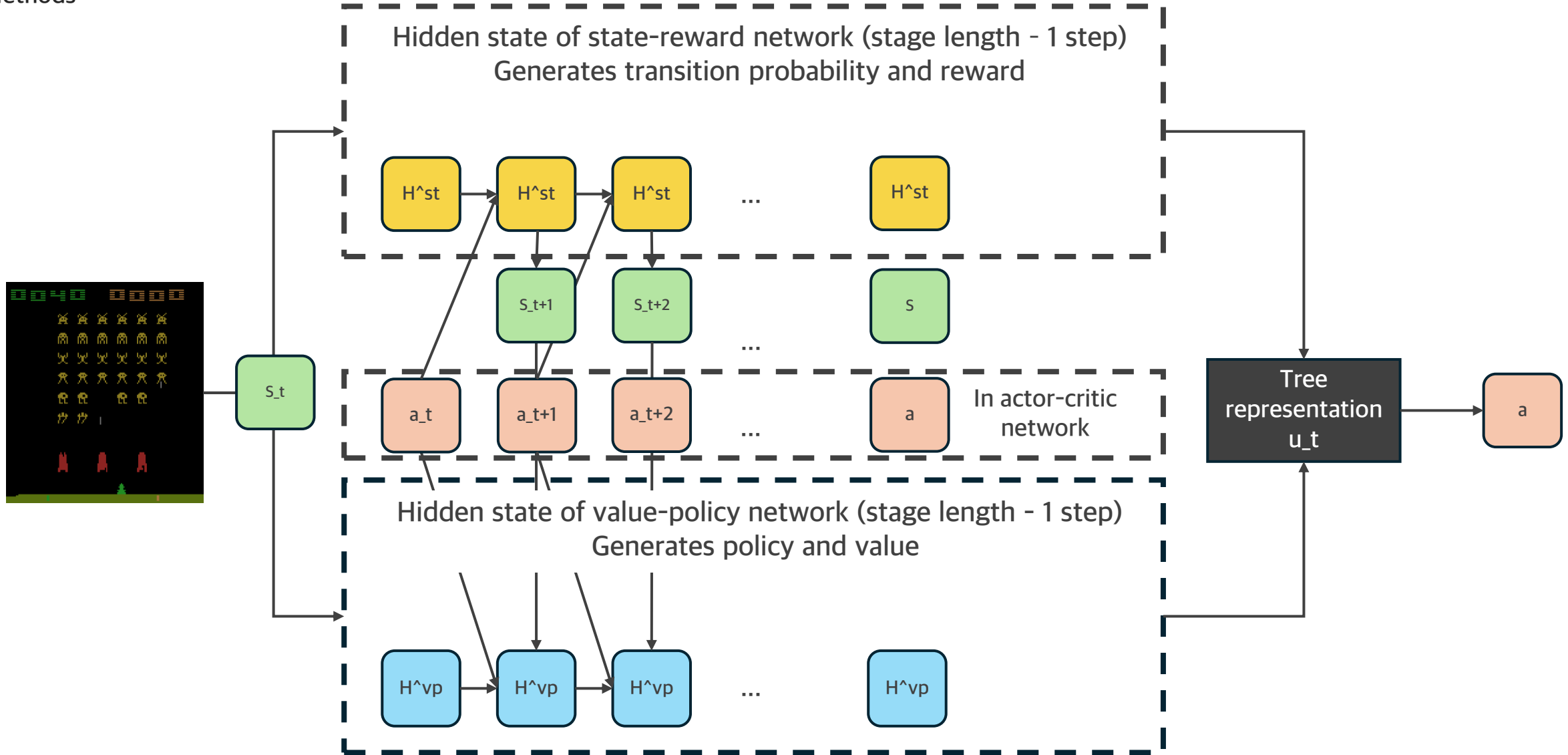
Methods



Thinker can straightforwardly reflect human planning:
How much we plan, How far we think and How we stop, reconsider and replan

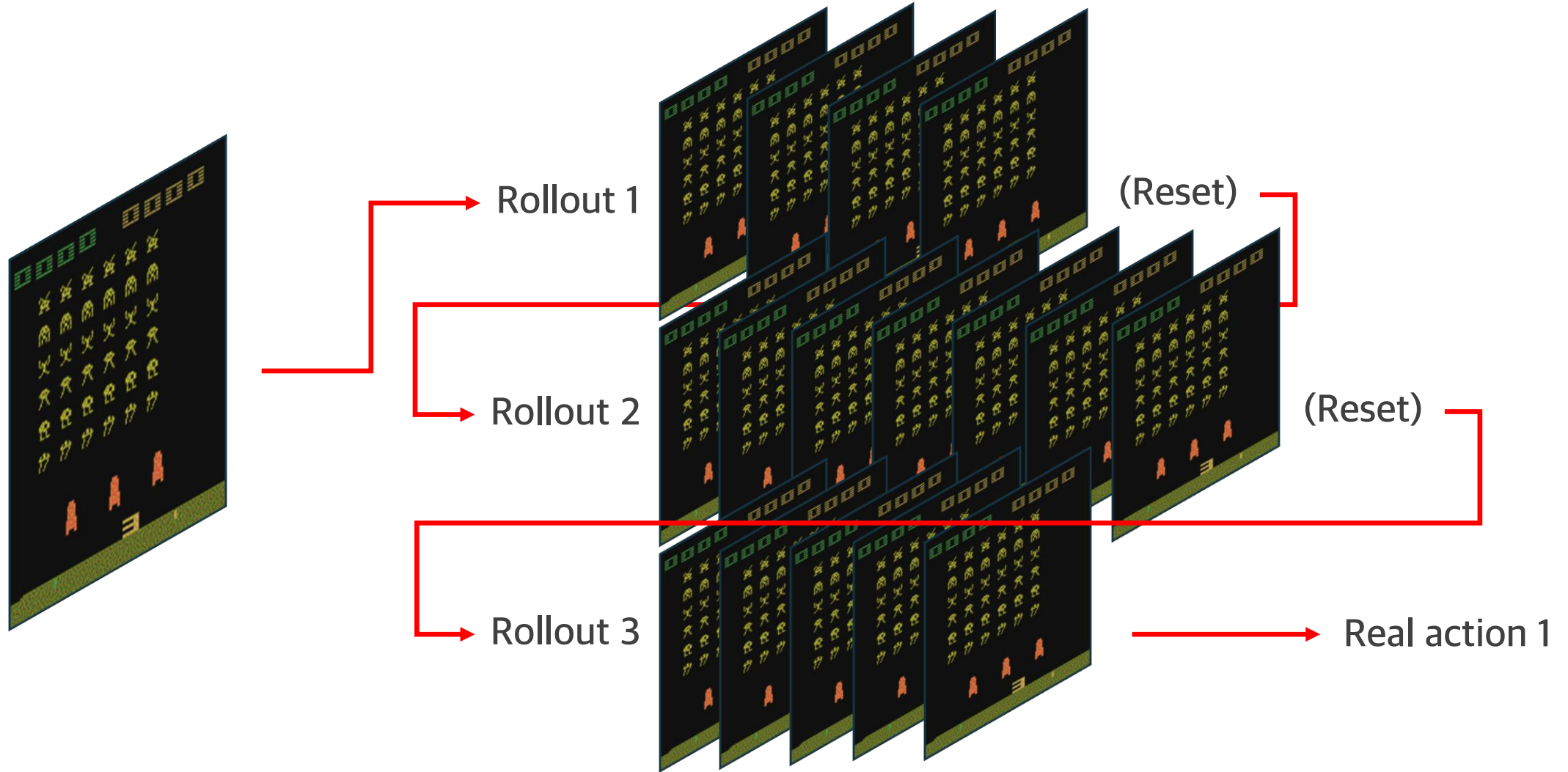
Algorithm of thinker

Methods



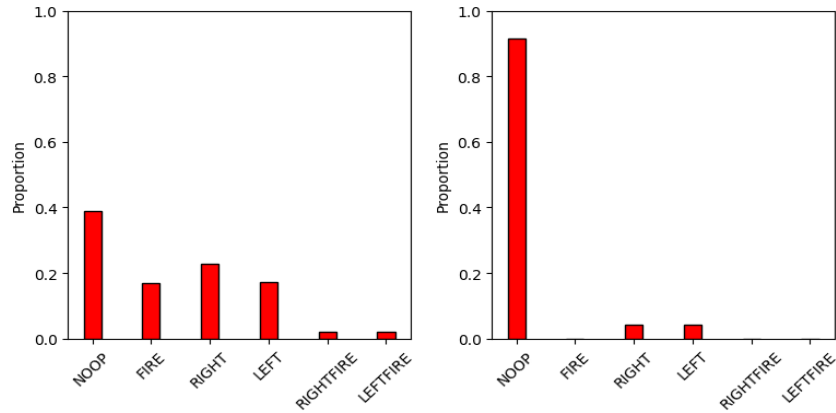
Visualization of dynamics in each state

Methods

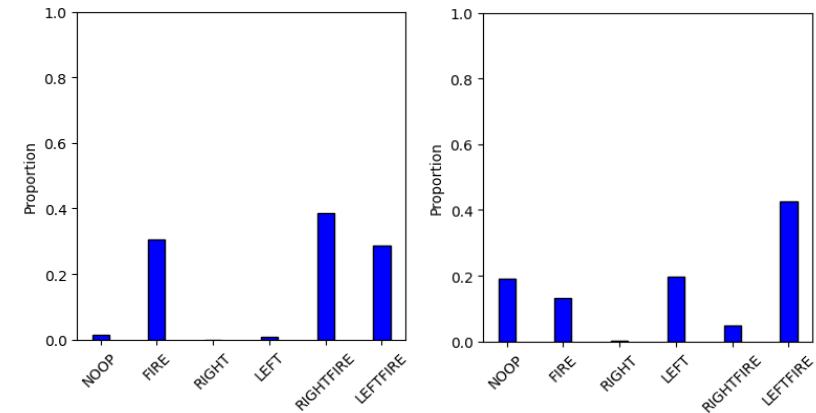


Why we need imitation learning before analysis

Methods

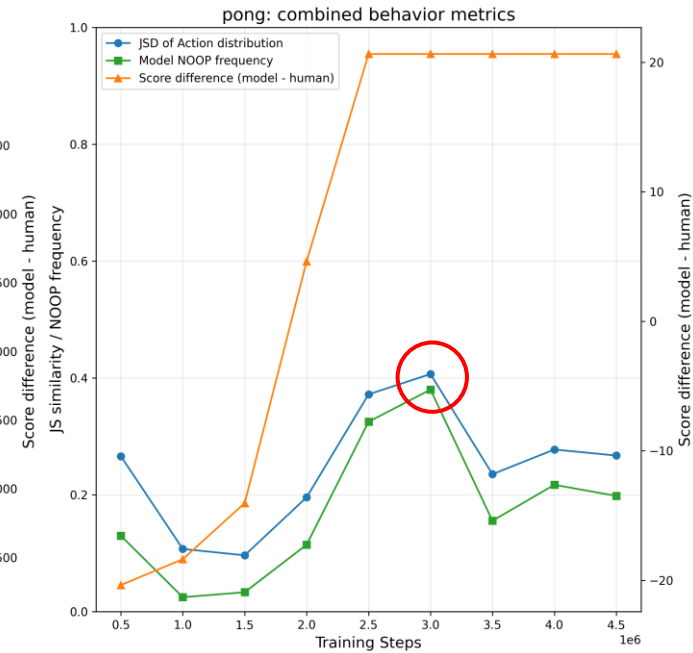
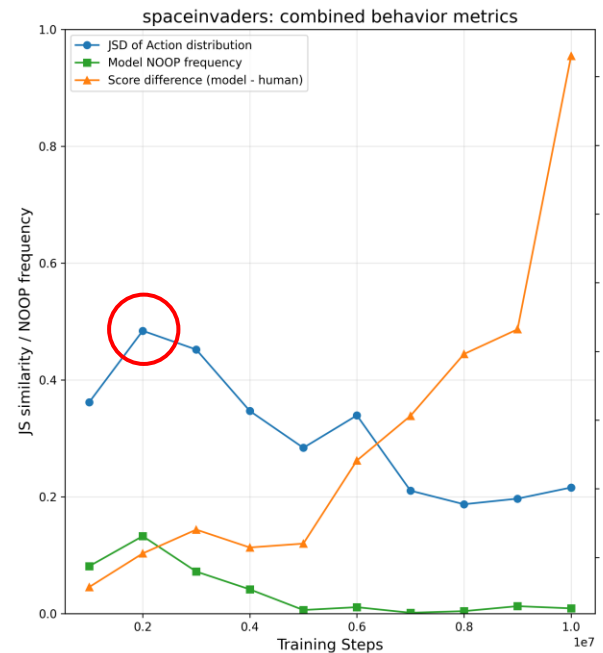


Action proportions for Space Invaders(L) and Pong(R) played by human



Action proportions for Space Invaders(L) and Pong(R) played by RL agent (Thinker)

Because human subject and RL model learn differently, we need to fit our model toward human subject.



Imitation learning details - step time weighting

Methods

Why we use both RL loss and supervised imitation learning loss?

→ our goal is to preserve the reward-driven gradient while constraining the policy to stay close to the human action distribution, so the model can achieve human-level performance without drifting away from human-like behavior.

For step time weighting, Let s be the aggregated planning time for transition, and λ is step time cost.

Weight is defined as following: $w_{t,b} = \frac{1}{1 + \lambda s_{t,b}^{time}}$

Final policy gradient loss and value loss are decided after multiplying step time weight.

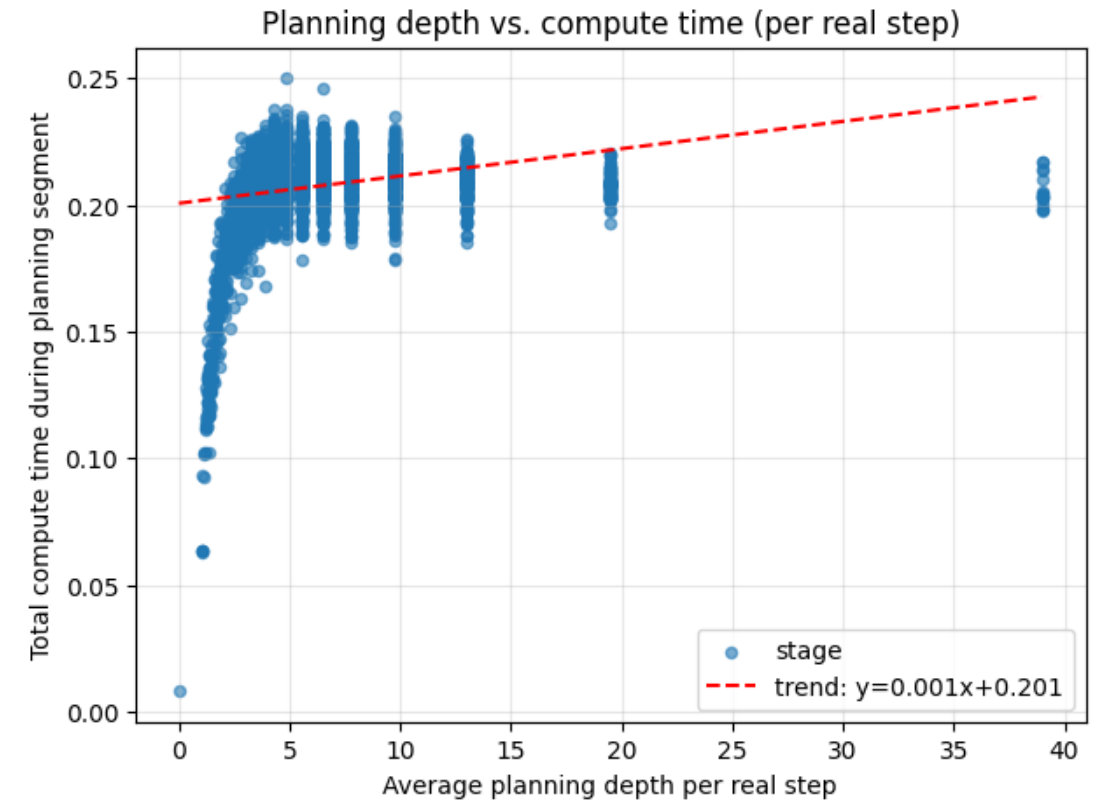
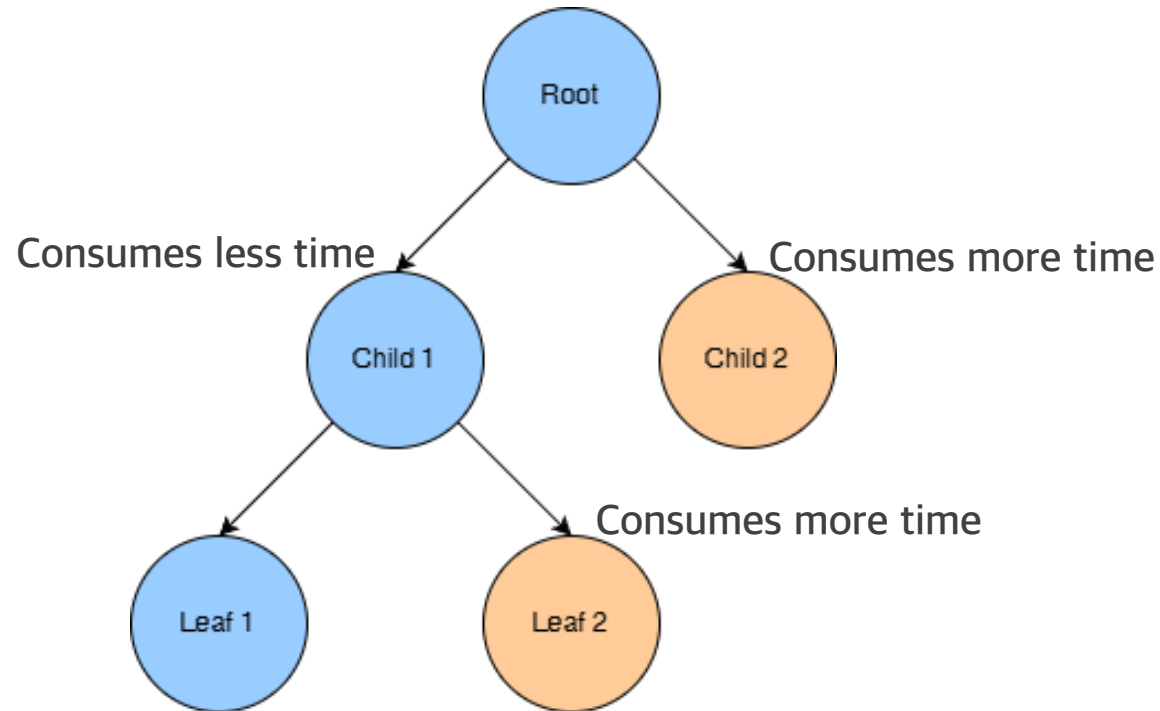
$$L_{pg}^{(k)} = \sum_{t,b} w_{t,b} m_{t,b}^{(k)} \left(-A_{t,b}^{(k)} \log \pi_{\theta} (a_{t,b} | s_{t,b}) \right), L_v^{(k)} = \sum_{t,b} w_{t,b} m_{t,b}^{(k)} \left(V_{\theta}^{(k)}(s_{t,b}) - \widehat{V}_{t,b}^{(k)} \right)^2$$

Imitation learning details - step time weighting

Methods

Blue nodes: nodes that agent has already visited

Orange nodes: nodes that agent hasn't visited



As a result, when the average depth increases, the probability of expanding new nodes typically increases as well, so computation time tends to rise

Imitation learning details - supervised learning

Methods

q is policy logit for each action, Margin loss enforces that the human action logit is larger than the logits of all other actions by at least a margin.

$$L_{margin} = \frac{1}{N} \sum_{i=1}^N \left(\max_a (q_i[a] + m_i[a]) - q_i[a_i^*] \right)$$

PVP pulls the human action logit toward +1, and if the agent's current greedy action differs from the human action, it pushes that greedy action logit toward -1.

$$L_{PVP} = \frac{1}{N} \sum_{i=1}^N \left((q_i^{human} - 1)^2 + d_i (q_i^{agent} + 1)^2 \right)$$
$$a_i^{agent} = \arg \max_a q_i[a], q_i^{human} = q_i[a_i^*], q_i^{agent} = q_i[a_i^{agent}], d_i = 1\{a_i^{agent} \neq a_i^*\}$$

Action difference loss is defined as standard cross entropy loss:

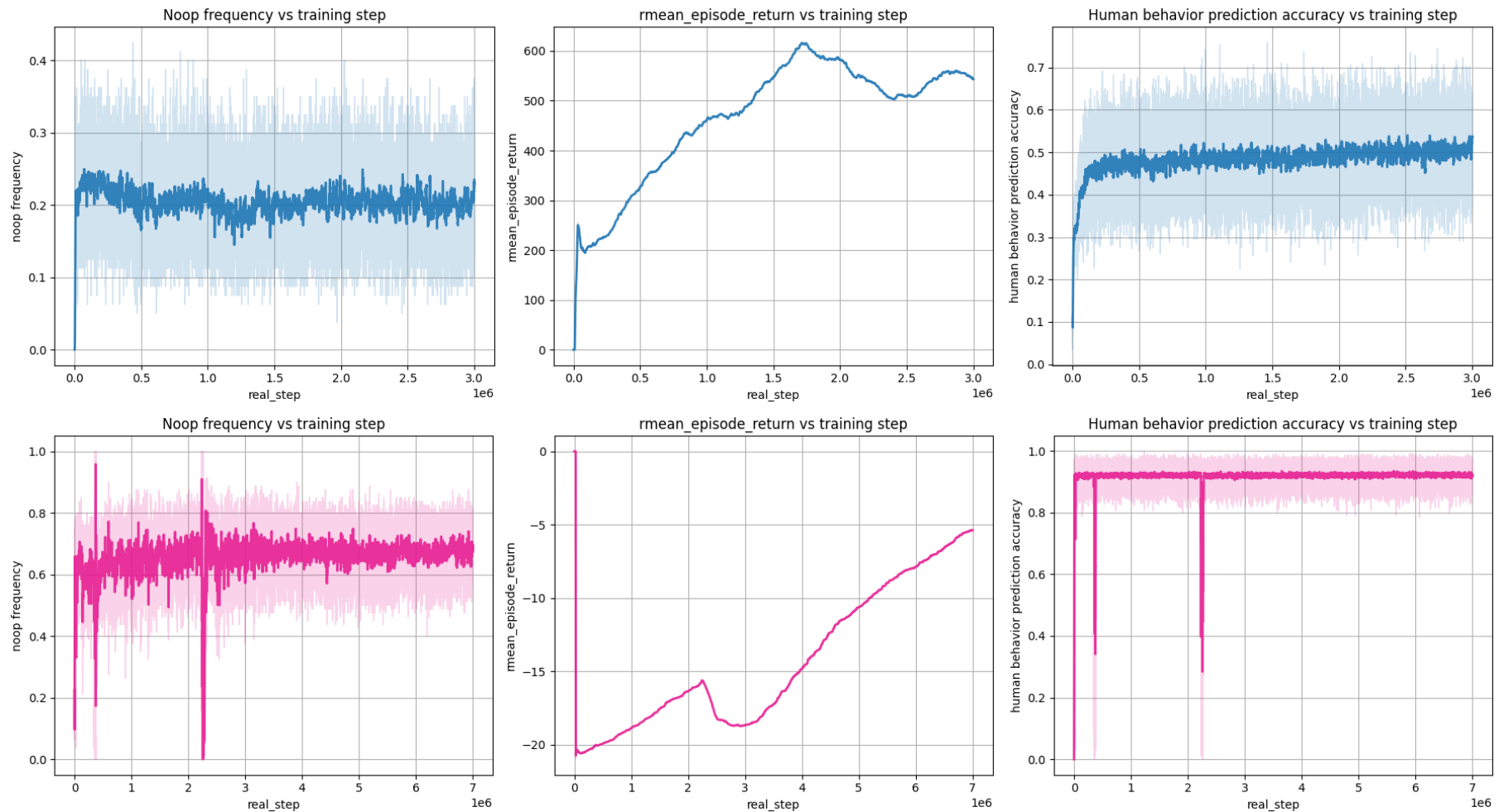
$$L_{diff} = -\frac{1}{N} \sum_{i=1}^N \log \pi_{\theta}(a_i^* | x_i)$$

This is total loss actor takes:

$$L_{actor,BC} = c_m L_{margin} + c_{pvp} L_{PVP} + c_{diff} L_{diff}$$

Imitation learning results

Results



NOOP usage remains at a nontrivial level throughout training rather than collapsing, while the model's performance steadily improves as reflected in the 50-episode average return. At the same time, the Thinker's action predictions on the human imitation dataset stay reliably above random chance (17%), indicating that the training procedure can increase score while preserving human-aligned action patterns.

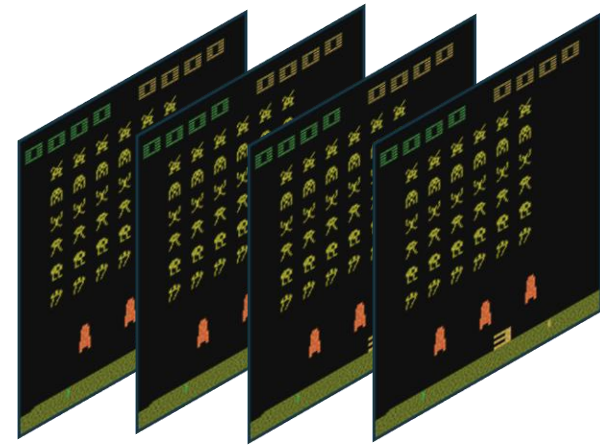
Regressor candidates

Methods

1. Average planning depth
 - Measure length of every rollouts
 - Example case: $(4+7+4)/3 = 5$
2. Action/Plan diversity
 - Quantifies how many distinct action trajectories are explored during rollouts
3. NOOP frequency
 - Compute proportion of actions that are NOOP within a given time window
4. Value Revision Error
 - Discrepancy between predicted value and real value
 - $VRE_i := |Q_i^{leaf,pre}(n_i^{leaf}) - Q_i^{leaf,post}(n_i^{leaf})|$
5. Step time
 - Elapsed total time from observation to action decision
6. Uncertainty
 - Policy entropy

Rollout 1

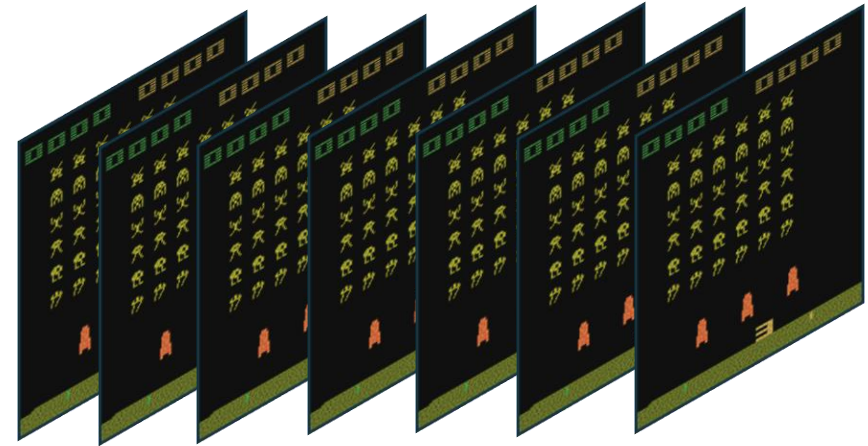
action sequence:
[0, 1, 2, 2, reset]



(Reset)

Rollout 2

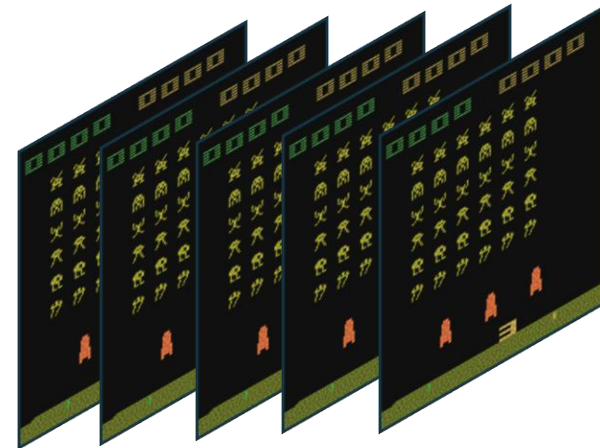
action sequence:
[0, 1, 4, 0, 3, 5, 2, reset]



(Reset)

Rollout 3

action sequence: [3, 3, 2, 2]



2. Research plan studying planning in the brain

My research question is...

“How does the brain allocate “thinking time” to predict futures and execute plans in real-time games?”

- ▶ Hypothesis 1: NOOP action tracks HC-mPFC planning coupling
- ▶ Hypothesis 2: Strategic NOOP is associated with HC-PFC planning networks
- ▶ Hypothesis 3: World model-HC representational alignment of task-state geometry

Continuous-Time Markov decision process (CTMDP)

Background

$$Sum_{MDP} = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] \quad Sum_{CTMDP} = \mathbb{E} \left[\int_0^{\infty} \gamma^t r(s(t), \pi(s(t))) dt | s_0 \right]$$

- ▶ For Continuous-Time Markov Decision Processes, decisions can be made at any time the decision maker chooses. In comparison to discrete-time Markov decision processes, continuous-time Markov decision processes can better model the decision-making process for a system that has continuous dynamics.
- ▶ Humans choose both an action and a decision time. the environment continues during deliberation, so waiting changes the state. We need a formalism where time is an explicit control variable, not an implicit index.
- ▶ A discrete-time MDP can be viewed as a time-discretized approximation of a CTMDP. Under this interpretation, the policy π selects both what to do and when to do it, making “waiting/NOOP” a legitimate decision rather than a modeling artifact.

NOOP actions: Key behavior of human game playing

Background

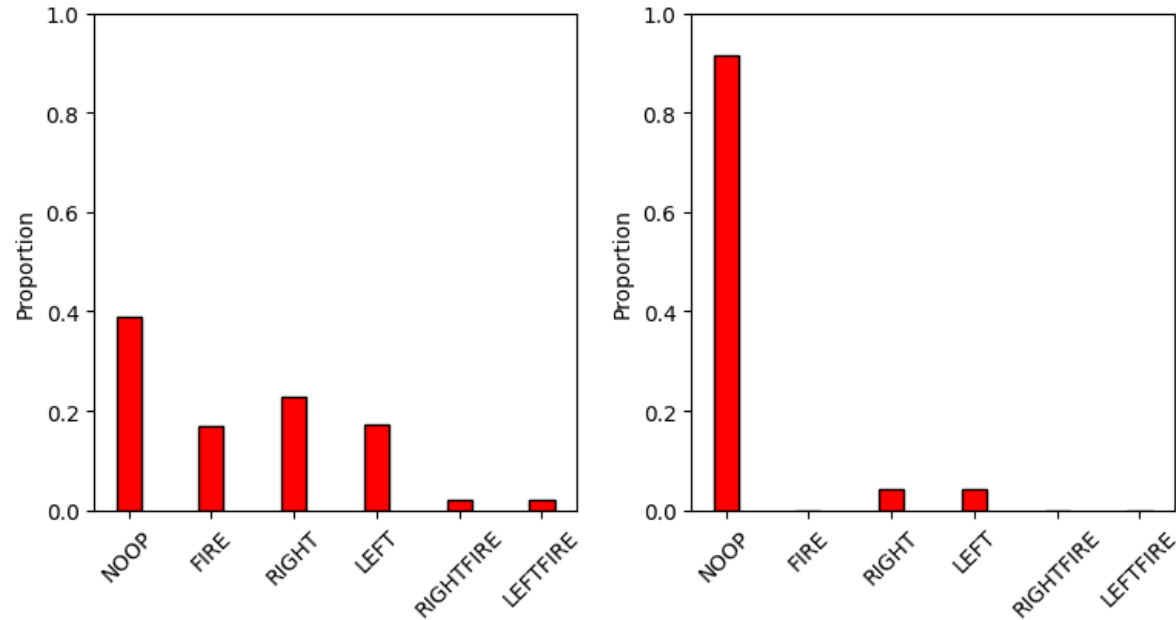


Fig 1. Action proportion for Space Invaders (L) and Pong (R) played by human

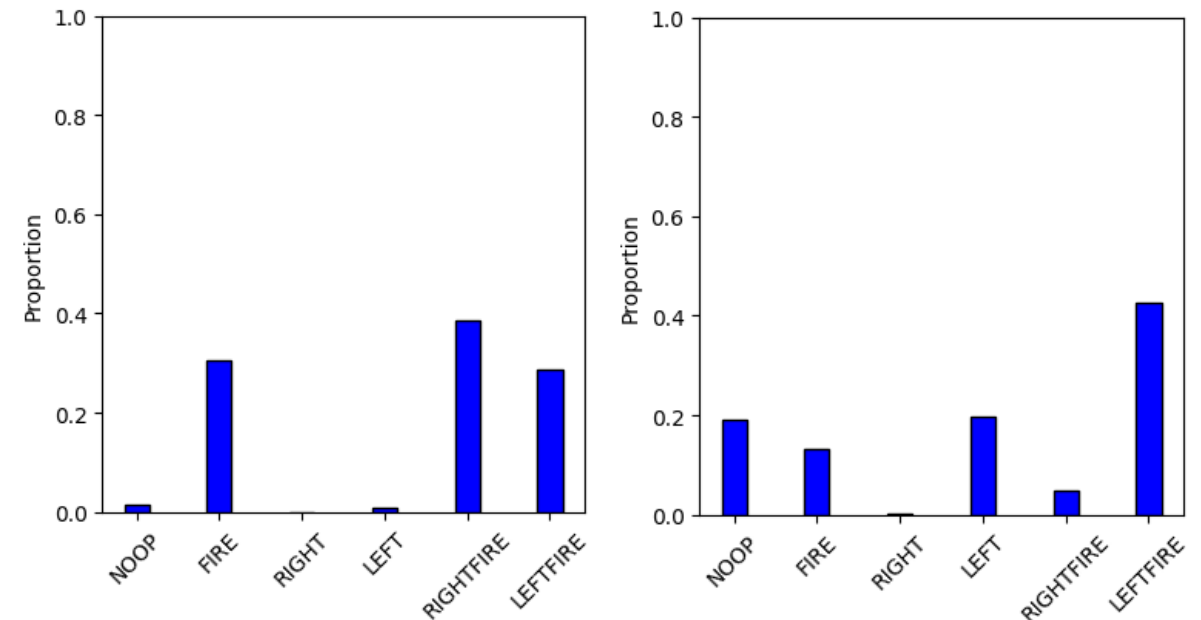


Fig 2. Action proportion for Space Invaders (L) and Pong (R) played by RL agent (Thinker)

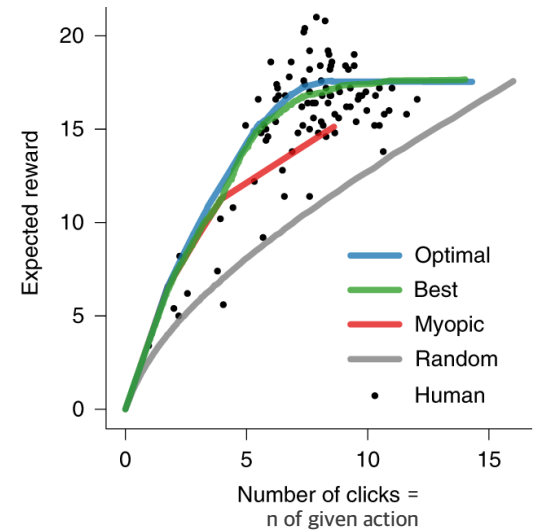
Humans exhibit **elevated NOOP probability** and different action entropy relative to the RL agent, suggesting distinct time-allocation and deliberation strategies

In our task, the environment continues during deliberation; thus NOOP (no-operation) captures thinking under ongoing dynamics. Two interpretations of NOOP are internal deliberation (buy computation time) timing and avoidance (wait for favorable dynamics). Our strategy is distinguishing them by testing planning-specific signatures: stronger hippocampus-mPFC coupling during NOOP.

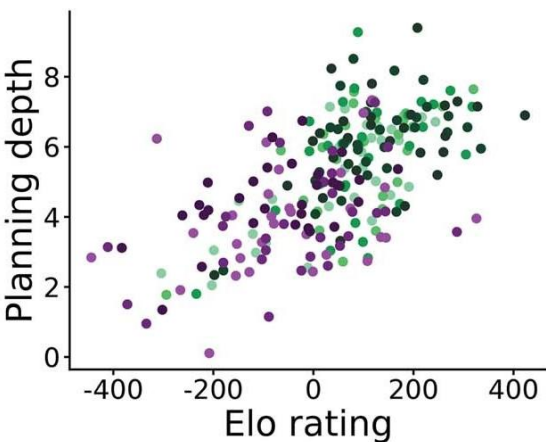
Behavioral Signatures of Human Planning

Background

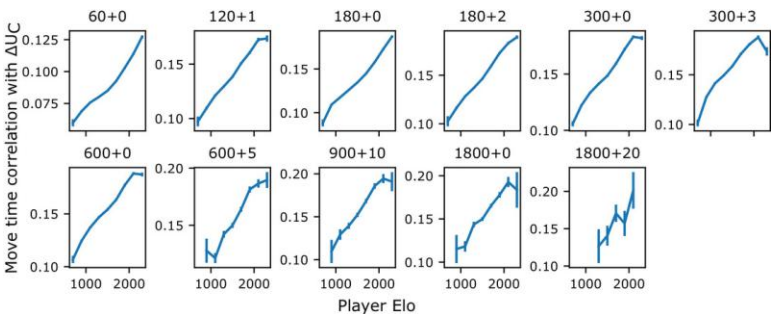
- ▶ Humans allocate planning effort adaptively, more thinking when the benefit of computation is higher, and less when costs dominate. (Callaway, 2022)
- ▶ Planning depth increases with expertise, and people adjust depth to the minimum required by the problem. (van Opheusden, 2023)
- ▶ In large-scale chess data, move times track the value of additional computation, especially in stronger players. (Russek, 2022)
- ▶ Planning and acting can be interleaved, consistent with continuous-time decision dynamics and “thinking as a control variable. (Nuzzi, 2025)



(L) Expected reward versus number of executed action. (Callaway, 2022)



(R) Elo rating versus computed planning depth in 4-in-a-row game. (van Opheusden, 2023)



(L) Graph for move time correlation with planning benefit(advantage from optimal move versus the move could be identified with less computation) between Elo rating, with different move time (Russek, 2022)

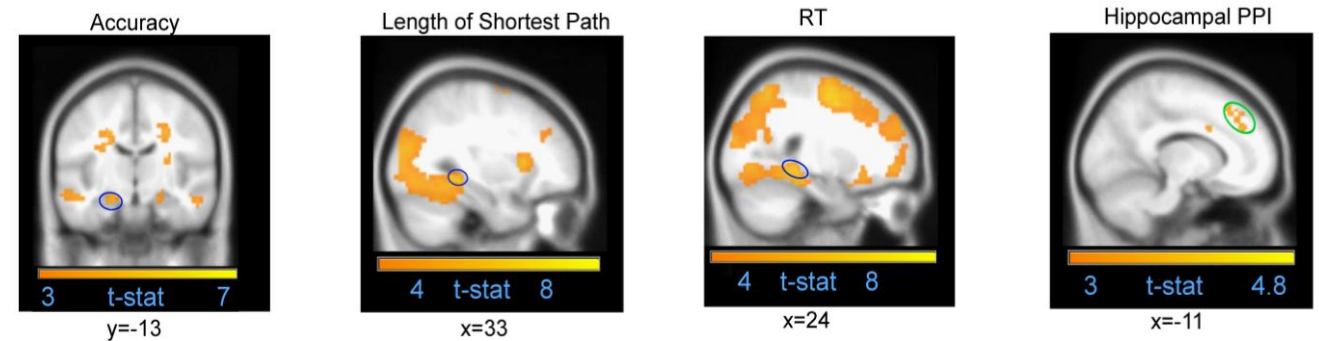
	Fixed effects					Random effects
	Estimate	Std Error	[0.025	0.095]	p-value	Variance
Intercept	20.143	0.131	19.886	20.401	< 0.001	0.544
Pre-planning time	-0.274	0.047	-0.366	-0.183	< 0.001	

(R) Results of the LMM examining the effect of pre-planning time on the length of the selected path in jumping game (Nuzzi, 2025)

Hippocampus-mPFC coupling tracks deliberation and correct choices

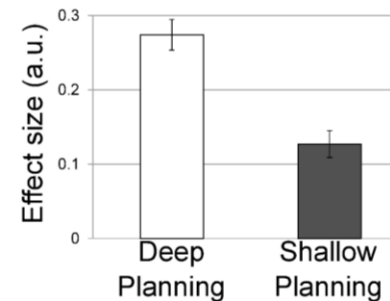
Background

- ▶ Hippocampal planning-phase responses correlate with subsequent choice accuracy and choice RT (reaction time), particularly in deep mazes require sequential choices with posterior hippocampal signals relating more to deliberation/RT and anterior hippocampal signals relating more to accuracy. (Kaplan, 2017)
- ▶ We treat NOOP epochs as a fine-grained proxy for reaction time (RT): NOOPs indicate longer state-level deliberation before committing to an action.
- ▶ Increased hippocampus-rd-mPFC coupling during sequential (“deep” vs. “shallow”) planning, and this coupling was higher preceding correct vs. incorrect choices (particularly within deep planning trials). (Kaplan, 2017)
- ▶ We hypothesize that Resting-state hippocampus-vmPFC intrinsic connectivity will affect transfer or generalization, suggesting trait-like variation that may shape subsequent task engagement.

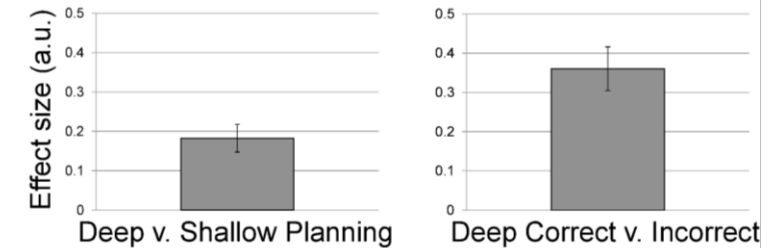


Hippocampal role in planning. (Kaplan, 2017)

Right Hippocampal RT Effect



Right Hippocampal PPI with rd-mPFC



(L) rd-mPFC voxels with stronger HC connectivity in deep vs shallow planning (effect size).

(R) HC-rd-mPFC connectivity is higher before correct than incorrect choices in deep planning. (Kaplan, 2017)

Hypothesis 1: NOOP tracks HC-mPFC planning coupling

Method

- ✓ Stronger hippocampus-mPFC BOLD functional connectivity will be associated with NOOP action.
- ✓ Crucially, this is not just a connectivity effect, The same moments should also express deeper model-based computation, giving a convergent behavioral-computational-neural signature of deliberation.
- ✓ Deeper Thinker's average planning depth will be associated with stronger neural activity in humans (e.g., hippocampus-mPFC/dIPFC)
- ✓ Coupling changes track reduced value revision error.

Hypothesis 1: NOOP tracks HC-mPFC planning coupling

Method

► NOOP Predicts HC-mPFC PPI

$$y_{mPFC}(t) = \beta_0 + \beta_{HC}S_{HC}(t) + \beta_{NOOP}P_{NOOP}(t) + \beta_{HC \times NOOP}(S_{HC}(t)P_{NOOP}(t)) + \beta_Z^T Z(t) + \epsilon(t)$$

$$PPI_{HC \times NOOP}(t) = S_{HC}(t) P_{NOOP}(t)$$

$$P_{NOOP}(t) = \frac{(I_{NOOP}(t) * h(t)) - \overline{(I_{NOOP} * h)}}{S_{HC}(t)}$$

$$S_{HC}(t) = BOLD_{HC}(t) - \overline{BOLD_{HC}}$$

► Functional Connectivity Tracks NOOP and Uncertainty at the State Level

$$FC_i^{HC-mPFC} = \gamma_N NOOP_i + \gamma_U Unc_i + \gamma_R VRE_i + \epsilon_i$$

Hypothesis 1: NOOP duration tracks HC-mPFC planning coupling

Method

- Deeper Thinker Planning Predicts Stronger Planning Activity in HC/mPFC/dIPFC

$$BOLD_i^{ROI} = \beta_{Depth} AvgDepth_i + \beta_U Unc_i + \beta_R VRE_i + \varepsilon_i$$

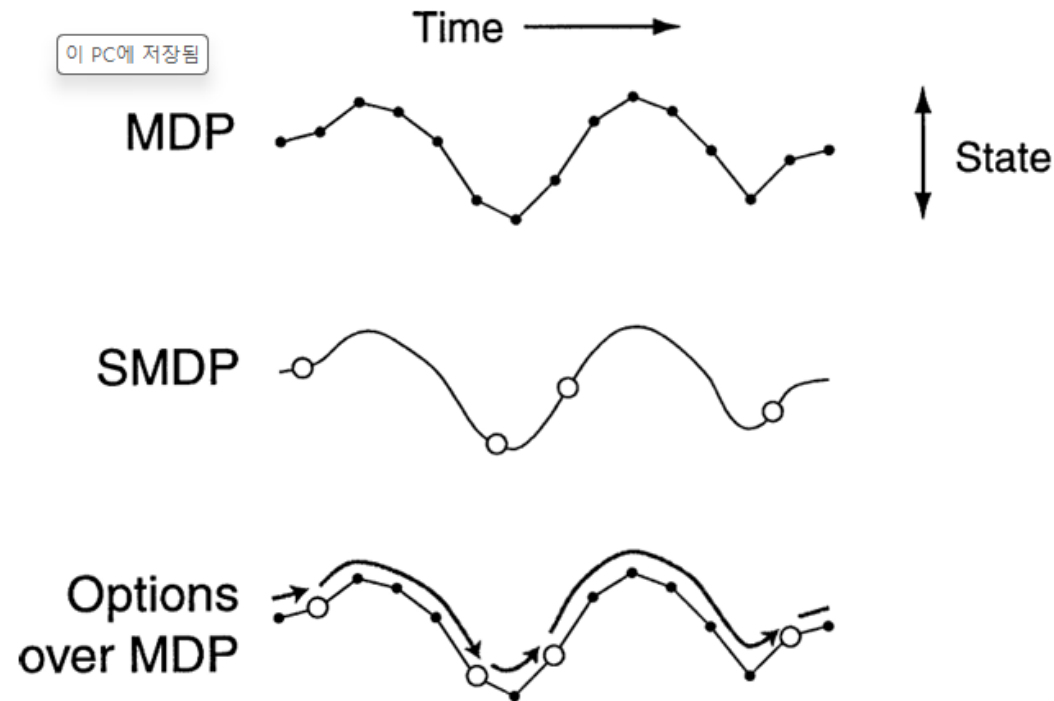
- Coupling Changes Track Reduced Value-Revision Error

$$VRE_i := |Q_i^{leaf,pre}(n_i^{leaf}) - Q_i^{leaf,post}(n_i^{leaf})|$$

$$\log(1 + VRE_{p,i}) = \beta_{0,p} + \beta_{PPI,p} PPI_{p,i} + \epsilon_{p,i}$$

Why a Continuous-Time MDP framework?

Background



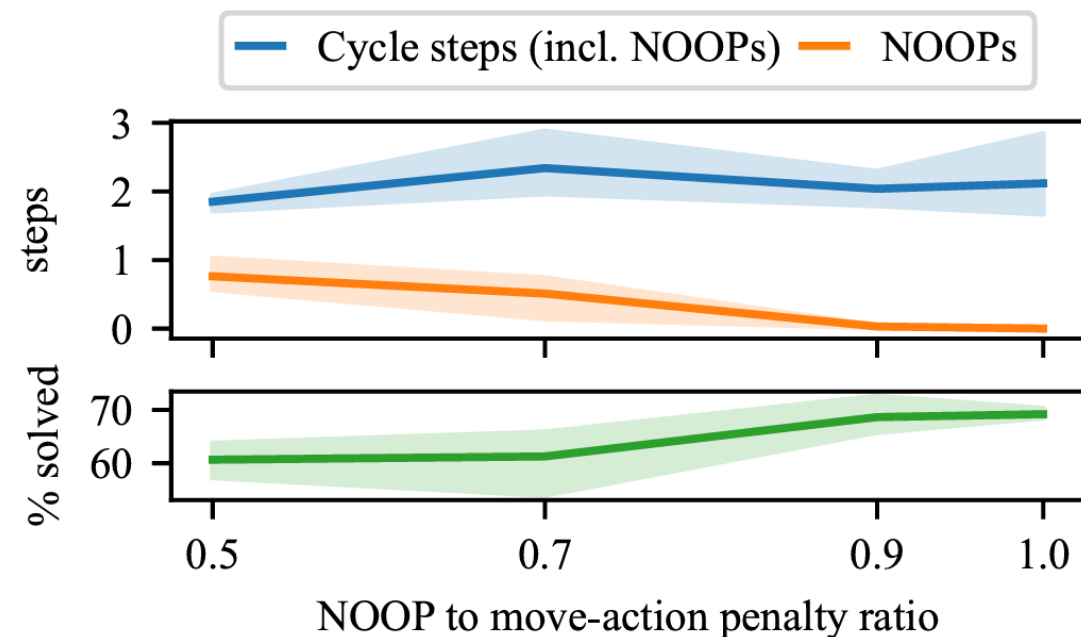
- ▶ In real tasks, agents choose not only what to do but when to act—so “waiting/thinking” is part of control, not noise.
- ▶ Options (SMDPs) capture variable-duration actions, but still treat time as chunked into discrete segments.
- ▶ This motivates a continuous-time (CTMDP) view, where decision time is a control variable and the agent can act at any moment.

Effects of NOOP actions in RL agent

Background

As the NOOP penalty decreased, the number of NOOP actions per episode increased. This indicates that the model leveraged the cheaper NOOP actions to acquire additional time needed for planning. (Taufeeque, 2025)

The total number of cycle steps per episode (including NOOP actions) remained roughly constant or slightly decreased. This suggests that while the agent acquired "thinking time" through the new means of NOOP actions, it tended not to excessively prolong the overall delay.



Proportion of cyclic steps according to NOOP penalty ratio (Taufeeque, 2025)

Hypothesis 2: Strategic NOOP is associated with HC-PFC planning networks

Method

- ✓ Participants who strategically use NOOP/pacing as extra “think steps” in high-uncertainty situations will show stronger hippocampus-PFC planning-network activation.

Hypothesis 2: Strategic NOOP is associated with HC-PFC planning networks

Method

- Strategic NOOP Sensitivity Index (SNI) Construction

$$\text{logit } Pr(NOOP_s = 1) = \alpha_p + \beta_{U,p} Unc_s, \quad SNI_p = \beta_{U,p}$$

- SNI Predicts HC-PFC Coupling

$$FC_{p,s}^{HC-PFC} = \alpha_p + \delta_{U,p} Unc_s + \beta_Z^T Z_{p,s} + \epsilon_{p,s} \text{ (1st level: within-subject)}$$

Estimate $\widehat{\delta_{U,p}}$ for each participant p

$$\widehat{\delta_{U,p}} = \gamma_0 + \gamma_{SNI} SNI_p + \eta_p \text{ (2nd level: between-subject link to SNI)}$$

- Decoding analysis with control comparison

$$FC_{p,s}^{HC-ROI} = \alpha_p^{HV} + \delta_{U,p}^{HV} Unc_s + \beta_Z^T Z_{p,s} + \epsilon_{p,s}^{HV}, \quad FC_{p,s}^{HC-PFC} = \alpha_p^{HP} + \delta_{U,p}^{HP} Unc_s + \beta_Z^T Z_{p,s} + \epsilon_{p,s}^{HP}$$

Background

-
- E**
- Heatmap showing the correlation of D and T for various combinations of (Fy, Fo, Hy, Ho) and (Fy, Fo, Hy, Ho). The color scale represents the number of steps, ranging from 0 (black) to 6 (gray). The diagonal is black, indicating zero steps for self-comparisons. The off-diagonal elements show varying degrees of gray, indicating the number of steps required for transitions between different combinations.
- F**
- Scatter plot showing the correlation of D and T for three conditions: POST, PERM, and NOISE. The y-axis is labeled 'Correlation D and T' and ranges from -0.3 to 0.1. The x-axis categories are POST, PERM, and NOISE. The POST condition shows a distribution of points centered around -0.15, with a green shaded area indicating a range. The PERM and NOISE conditions show distributions centered around -0.05, with gray shaded areas indicating a range.

A

HPC off-task sequences (β)

HPC on-task decoding (%)

B

OFC off-task sequences (β)

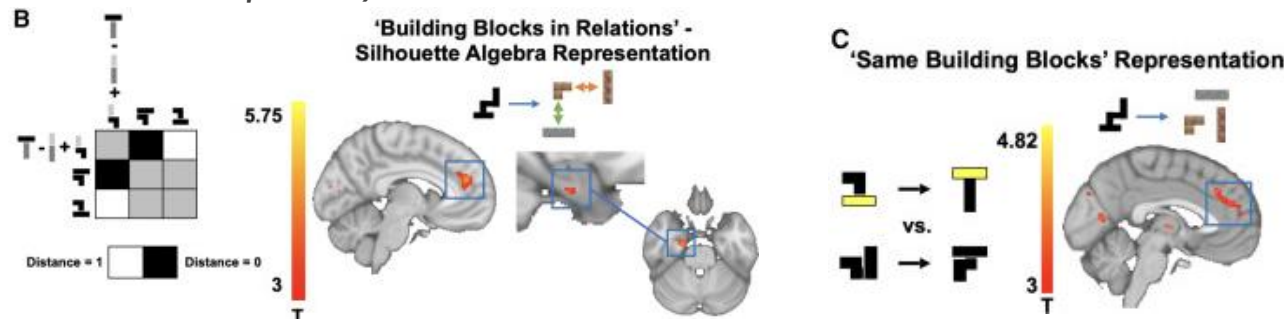
OFC on-task decoding (%)

C

Decoding/Error relation (β)

HPC OFC

- Replay in the hippocampal-prefrontal circuit can be generative and compositional: during compositional problem solving, activity reflects assembling building blocks into candidate hypotheses, and replay sequences become progressively refined toward the correct composition. (Schwartenbeck, 2023)



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Hypothesis 3: World model-HC alignment of task-state geometry

Method

- ✓ Replay studies suggest that hippocampus carries a structured task-state geometry; here I test whether that geometry aligns with Thinker's world model hidden-state representation.
- ✓ I treat Thinker's hidden-state embedding h_t as a candidate model of task-state representation, and compare it to hippocampal multivoxel patterns.
- ✓ I will quantify alignment using RSA between Thinker and hippocampal RDMs, testing whether pairs of moments that are close in Thinker space are also similar in hippocampal patterns.
- ✓ To check that any alignment is not driven by low-level visual similarity, I will compare against V1 and compute partial RSA controlling for V1; I will also compare against PFC to ask whether the alignment is hippocampus-specific or distributed across a planning network.

Hypothesis 3: World model-HC alignment of task-state geometry

Method

- Construct RDMs (World model vs ROI patterns)

$$t \in \{1, \dots, T\}, t' \in \{1, \dots, T\}, \text{ with } t' \neq t$$
$$D_r(t, t') = 1 - \text{corr}(x_t^{(r)}, x_{t'}^{(r)}), D_{Thinker}(t, t') = |h_t - h_{t'}|_2,$$

- RSA correlations (define a, b, c)

$$a = \text{corr}(u(D_{HC}), u(D_{Thinker})), b = \text{corr}(u(D_{HC}), u(D_{V1})), c = \text{corr}(u(D_{Thinker}), u(D_{V1}))$$

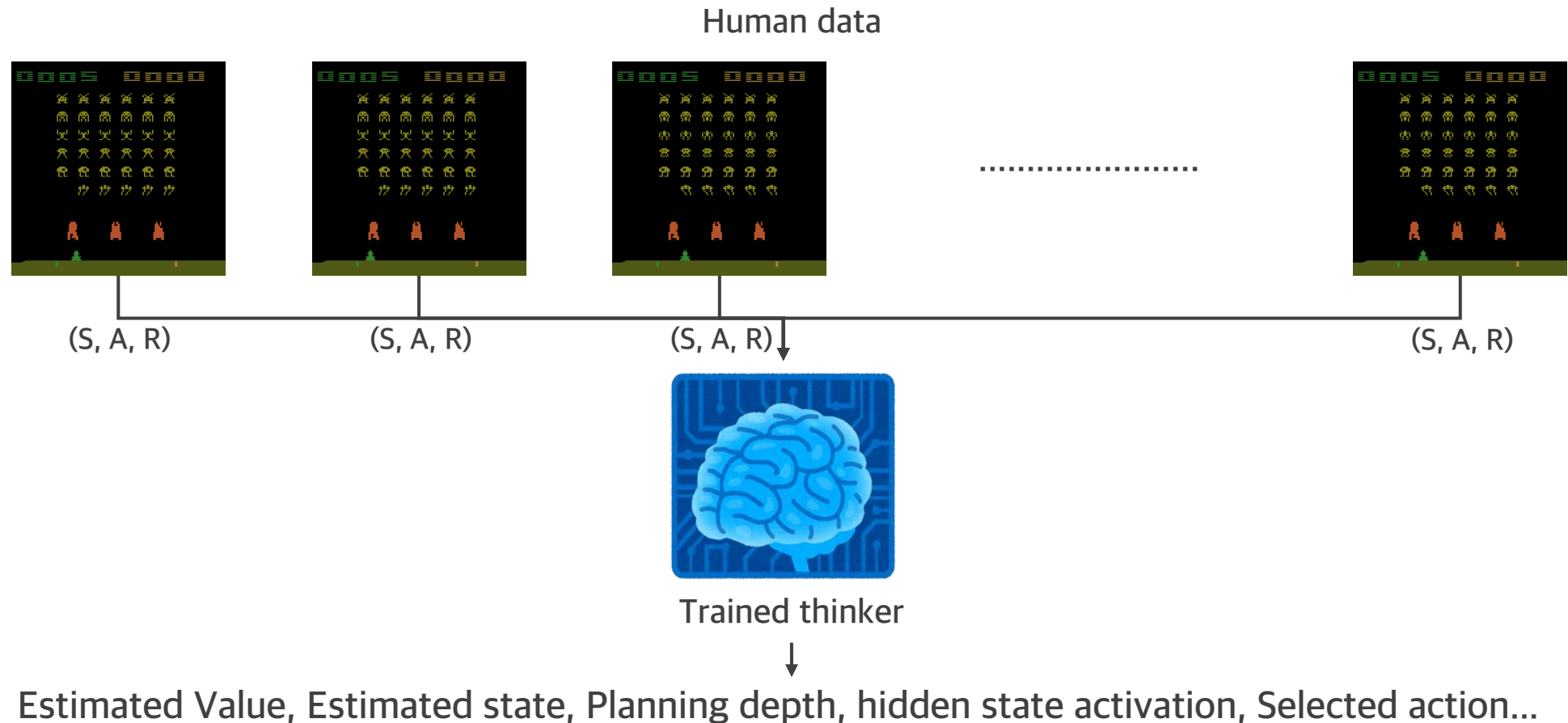
- Partial RSA controlling for V1 (visual confound removed)

$$pRSA_{HC,V1} = (a - bc) \left((1 - b^2)(1 - c^2) \right)^{-1/2}$$

Future plan

Imagination data generation method

- 1) All data for analysis will be generated with human behavioral data by feeding each (S, A, R) pair to Thinker model
- 2) Collect search result, model activation for each step



Future plan

Future direction

- 1) Train Thinker with each subject, Generate imagination data for human behavioral data, Test three hypotheses
- 2) Ablation analysis for loss function used training Actor and World model
- 3) Train other domains of game with open dataset (ex. Super mario)

Closing remarks

A central problem in cognitive neuroscience is that planning and internal evaluation are mostly latent, and behavior alone rarely reveals when people are truly simulating futures.

By pairing real-time game behavior with a world-model-based computational proxy, We can recover moment-by-moment latent variables and test whether they align with distributed brain networks and representational geometry across regions.

If it generalizes across people and tasks, this becomes a scalable bridge from behavior to brain, and a route toward brain-aligned computational models.

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Thank You

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Appendix

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Neural evidence brain updates structured information during game playing

Background

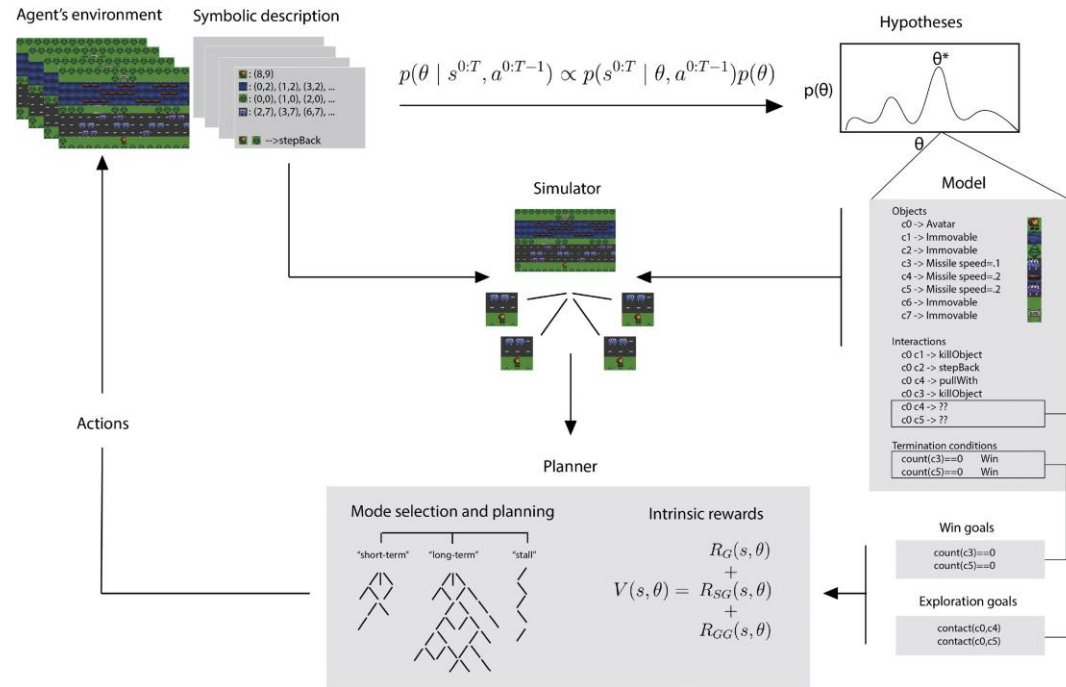
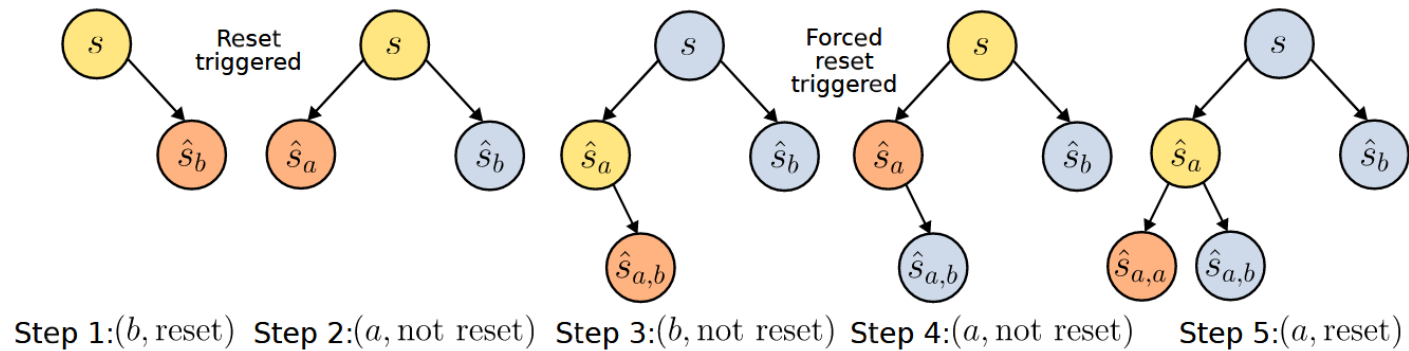
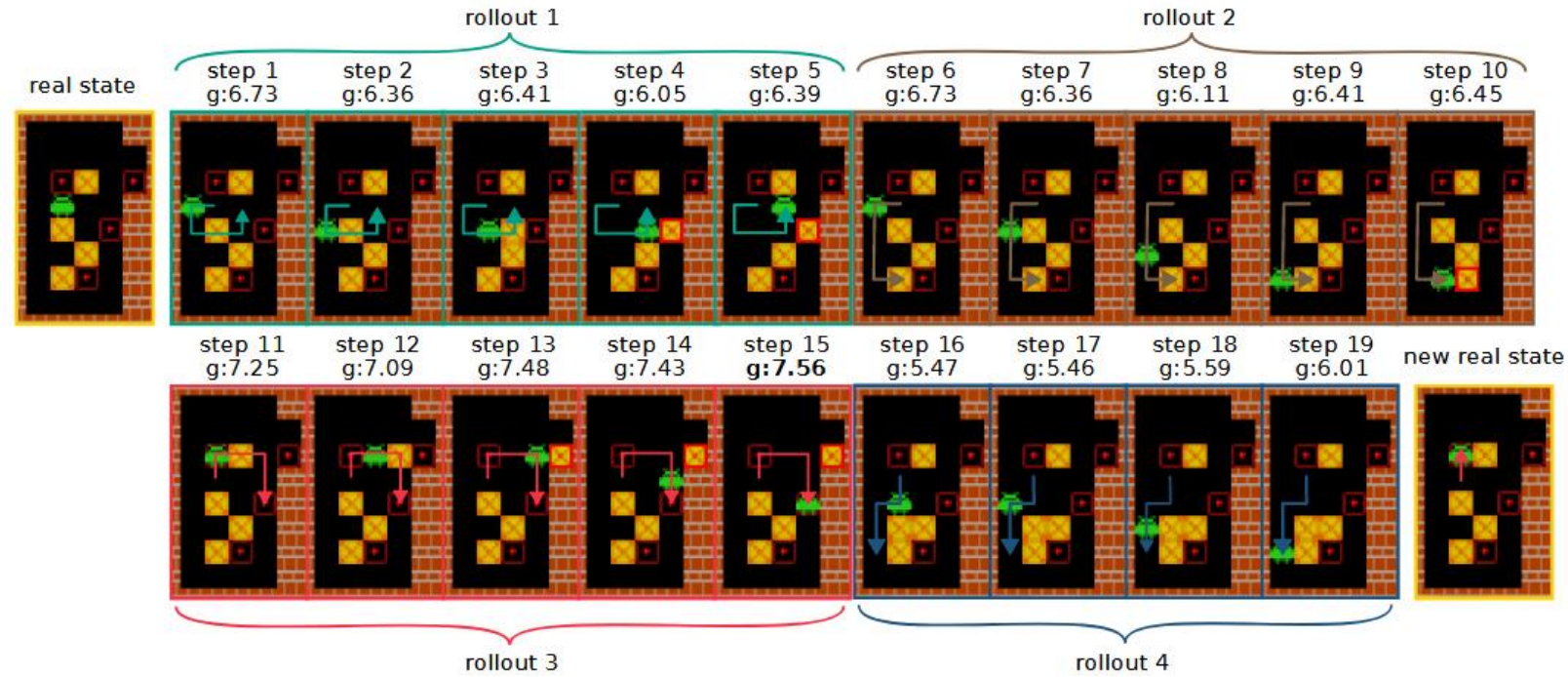


Fig 11. Visual description of EMPA (the Exploring, Modeling, and Planning Agent) model.

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```

Thinker: Model-Based RL agent that *explicitly plans to act*.

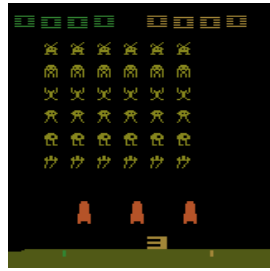
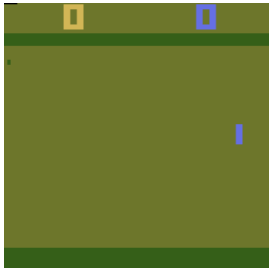
Research plan



Chung, S., Anokhin, I., & Krueger, D. (2023). Thinker: Learning to plan and act. Advances in Neural Information Processing Systems, 36, 22896-22933.

Method:

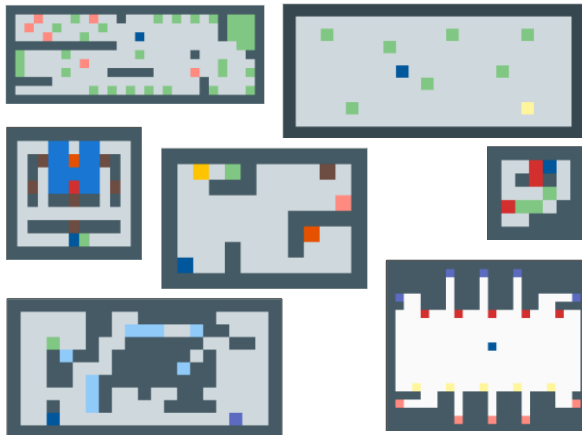
Research plan



Train Thinker
for each game

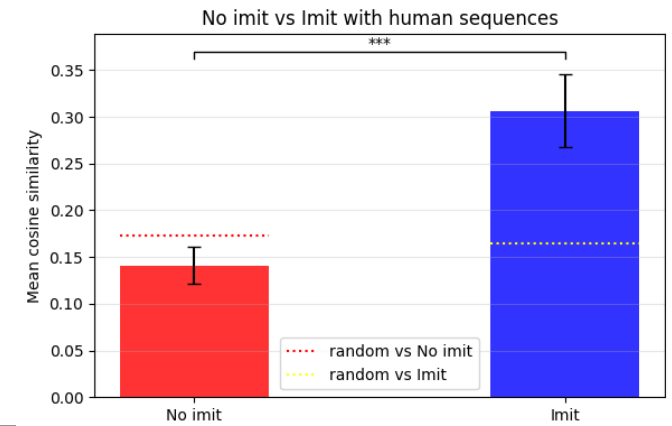
Find underlying
characteristics

Find ROI where our data can
explain voxel activity



* Each game consists of 9 levels (from lvl 0 to lvl 9)

Appendix





Templates

YYYY.MM.DD.

Jeongmin Seo

0. Result

WRYYYYYYY

*PCC: Pearson Correlation Coefficient

Main text: 20pts

Figure caption: 12pts

Supplementary text: 16pts

- Consider modality of your data:
 - Most of recent DL models are well suited with image/language data
 - Unique modality makes it harder (ex. fMRI, NIRS, EEG...)
 - Try implementing proper encoder/simple model

Human should look ahead at least 8 steps ahead to avoid wrong path: Designed several maze requires different planning depth